

IMO 2019 Exercise 1

1. If the eight digits 1, 2, 3, 4, 5, 6, 7, 8 are randomly permuted, what is the probability that the resulting eight-digit number is divisible by 8?
2. How many ways are there to arrange 10 identical red balls, 5 identical green balls and 5 identical blue balls in a row so that no two adjacent balls are of the same colour?
3. There are three identical red balls, three identical yellow balls and three identical green balls. In how many different ways can they be split into three groups of three balls each?
4. Eight couples attend a gathering. They are to be divided into four groups so that each group consist of two men and two women, and that no man may be in the same group as his wife. How many different ways of grouping are there?
5. A number is *good* if the sum of its digits is 18, (for example 130518 is a good number), find the number of good 6-digit numbers.
6. Let X be a randomly chosen subset of $\{1, 2, \dots, 2017\}$. A positive integer n is said to be 'good' if both n and $n + |X|$ are elements of X (where $|X|$ denotes the number of elements of X). Find the expected value of the number of 'good' integers.
7. A robot starts from $(0,0)$, it moves up, down, left or right a unit length each time. How many ways from $(0,0)$ to $(0,0)$ in $2n$ steps?
8. Let n be a positive integer and suppose a $2n \times 2n$ grid is given. Suppose we color exactly $2n^2$ square boxes of the grid in such a way that the following condition is satisfied: If a box is colored then none of the boxes which share only a vertex with that box is not colored. How many ways to color the grid? Two configurations are distinct even if one can be transformed to the other by rotation or reflection.
9. Let $1 \leq r \leq n$ and consider all subsets of r elements of the set $[n] = \{1, 2, \dots, n\}$. Each of these subsets has a smallest member. Let $F(n, r)$ be the arithmetic mean of these smallest numbers. Show that $F(n, r) = \frac{n+1}{r+1}$.

10. Denote by $P_n(k)$ the number of permutations of $1, 2, \dots, n$ which have k fixed points. Show that $\sum_{k=0}^n kP_n(k) = n!$.
11. 16000 committees were formed with members coming from 1600 councillors. Each committee consisted of exactly 80 members. Show that there exist two committees, and they had at least 4 common members.
12. In a contest there are m candidates and n judges, where $n \geq 3$ is an odd integer. Each candidate is evaluated by each judge as either pass or fail. Suppose that each pair of judges agrees on at most k candidates. Prove that $\frac{k}{m} \geq \frac{n-1}{2n}$.

Notes: We now consider a problem of slightly different flavor. Consider sequences of length n and such that each sequence contains only 0s or 1s. Define the “distance” between two such sequences as the number of positions they differ. For example 1101011 and 1011000 are two sequences of length 7, and they differ on positions 2, 3, 6 and 7, hence the distance between them is 4. Using set language, we say that a set contains 7 elements 1, 2, 3, 4, 5, 6 and 7, as sequence 1 are non-zero at positions 1, 2, 4, 6, and 7, and we may consider sequence 1 as a subset containing elements 1, 2, 4, 6 and 7. Likewise sequence 2 is a subset containing 1, 3 and 4. Elements belong to sequence 1 or sequence 2, but not both, in this case consist of 2, 3, 6 and 7, together form a subset, called the “symmetric difference” of the two subsets. One can verify easily that the distance between the sequences corresponds precisely to the number of elements in the symmetric difference. Now our concern is, given n and a restriction on the distances between the sequences, at most how many sequences we can get?

13. Given m sequences of length n and the sequences contain only 0s and 1s. Suppose the distance between any two such sequences is at least d , then $m \leq \frac{2d}{2d-n}$.
14. Let k be a positive integer. Find the number of non-negative integers n less than or equal to 10^k satisfying the following conditions:
- (i) n is divisible by 3;
 - (ii) Each decimal digit of n is one of the digits 2, 0, 1 or 7.
15. Let n be a positive integer. A permutation $\{x_1, x_2, \dots, x_{2n}\}$ of the set $\{1, 2, \dots, 2n\}$ has the property P if $|x_i - x_{i+1}| = n$ for at least one i . Show that for every n , $n \geq 1$,

the number of permutations with the property P is greater than the number of permutations which don't have.

20 July 2018

Easy

1. If the eight digits 1, 2, 3, 4, 5, 6, 7, 8 are randomly permuted, what is the probability that the resulting eight-digit number is divisible by 8?

It suffices to consider the last three digits. There are $8 \times 7 \times 6 = 336$ possibilities, and we need to count how many of these are divisible by 8. To be divisible by 8, the unit digit must be 2, 4, 6 or 8. We consider these cases one by one.

- If the unit digit is 2, the tens digit must be 1, 3, 5 or 7 in order for the number to be divisible by 4. If the tens digit is 1 or 5, there are 3 choices of hundreds digit (e.g. if the tens digit is 1, then the hundreds digit must be 3, 5 or 7 as 312, 512 and 712 are divisible by 8); if the tens digit is 3 or 7, there are 4 choices (namely, 2, 4, 6, 8) Hence there are $3 + 3 + 4 + 4 = 14$ possibilities in this case. **(Some Errors here?)**
- If the unit digit is 4, the tens digit must be 2, 6 or 8, leading to 8 possibilities, namely, 624, 824, 264, 864, 184, 384, 584, 784.
- The case where the unit digit is 6 is essentially the same as the first case; 14 possibilities.
- The case where the unit digit is 8 leads to 10 possibilities (it is slightly different from the second case), namely, 128, 328, 528, 728, 168, 368, 568, 768, 248 and 648.

It follows that the answer is $\frac{14+8+14+10}{336} = \frac{23}{168}$. **(Some Error Here)**

2. How many ways are there to arrange 10 identical red balls, 5 identical green balls and 5 identical blue balls in a row so that no two adjacent balls are of the same colour?

There must be at least 1 non-red ball placed between two red balls. Hence a green ball and a blue ball can be placed next to each other at most once. In the case no two non-red balls are placed next to each other, the green balls and the blue balls must be placed between the red balls alternately. If a red ball is placed first in the left most position, then there are $\frac{10!}{5!5!} = 252$ ways of doing that. If a red ball is placed in the

right most position, again there are $\frac{10!}{5!5!} = 252$ ways of doing that. Together there are 1008 ways of arranging the balls. If there is a blue ball and a green ball placed next to each other, it is regarded as a “combined” ball. With the other 4 blue balls and 4 green balls, they must be placed between the red balls. There are $\frac{9!}{1!4!4!} = 630$ ways of arrangements. By swapping the relative positions of the green and blue ball of the “combined ball”, there are again 630 ways. Together we have $504 + 1260 = 1764$ ways of arrangement.

3. There are three identical red balls, three identical yellow balls and three identical green balls. In how many different ways can they be split into three groups of three balls each?

Let R, Y, G denote red, yellow and green ball respectively. Call a group ‘pure’ if all balls are of the same colour. We have the following cases:

- 3 pure groups — there is only 1 way of grouping
- 2 pure groups — this is not possible (if two groups are pure, so is the third)
- 1 pure group — there are 3 ways of grouping, corresponding to choosing one colour (say, R) out of three to form a pure group (the other two groups must be YYG and YGG)
- no pure group — either every group contains R (2 ways, {RYG, RYG, RYG} or {RYG, RYY, RGG}), or otherwise the grouping is {RRX, RXX, XXX} where each X is either Y or G; there are two choices for the first X, then two choices for the ‘XX’ in ‘RXX’ (e.g. after RRY we can choose RYG or RGG for the second group, but not RYY), giving a total of $2 + 2 \times 2 = 6$ ways in this case.

Hence the answer is $1 + 0 + 3 + 6 = 10$.

4. Eight couples attend a gathering. They are to be divided into four groups so that each group consist of two men and two women, and that no man may be in the same group as his wife. How many different ways of grouping are there?

We first divide the 8 men into 4 groups of two. Pick any man A and there are 7 ways to choose his groupmate B . Then pick any other man C and there are 5 ways to choose his groupmate D . Next pick any one of the remaining men E and there are 3 ways to choose his groupmate F . Finally the last two men G and H must be in the same group. Hence there are $7 \times 5 \times 3 = 105$ ways to group the men. Now we number the groups 1 to 4 (i.e. A and B are in Group 1, and so on), and let $a, b, \dots, h$ be the wives of $A, B, \dots, H$ respectively. We consider two cases.

Case 1: The husbands of the two women in Group 1 are in the same group

There are 3 ways to choose two women into Group 1. WLOG assume c and d are put into Group 1. Then the following groupings are possible according to the positions of a and b :

- If a and b join the same group (3 possibilities), the grouping is fixed. (For example, if both a and b join Group 3, then since g and h cannot join Group 4 they are both forced to join Group 2.) Hence there are 3 arrangements in this case.
- If a joins Group 3 and b joins Group 4 (or vice versa, leading to 2 possibilities), there are 2 choices for each of the remaining place in Group 3 (for g or h) and Group 4 (for e or f), giving a total of $2 \times 2 \times 2 = 8$ arrangements in this case.
- Otherwise, one of a and b joins Group 2, the other joins either Group 3 or Group 4 (4 possibilities). Let's say a joins Group 2 and b joins Group 3. Then e and f must join Group 4 but g and h are free with 2 choices, giving $4 \times 2 = 8$ arrangements in this case.

Altogether, there are $3 \times (3 + 8 + 8) = 57$ arrangements in Case 1.

Case 2: The husbands of the two women in Group 1 are in different groups

There are $C_2^6 - 3 = 12$ ways to choose two women into Group 1. WLOG assume c and e are put into Group 1. Then the following groupings are possible according to the positions of d and f :

- If d and f are in the same group, they must be in Group 4. Then a, b, g, h are free to join Group 2 and Group 3, so there are $C_2^4 = 6$ possible arrangements.
- If neither d nor f go to Group 4, then d and f must be in Group 3 and Group 2 respectively. Hence there are only 2 arrangements, with a and b joining Group 4 and g and h taking up the remaining two places in either way.
- Otherwise there are 2 ways for one of d and f to go to Group 4, say, d joins Group 4. Then f must be in Group 2. There are 3 possible arrangements for g and h (both join Group 3, or choose one of them to join Group 3 and the other to join Group 2), and then 2 possible arrangements for a and b . Hence there are totally $2 \times 3 \times 2 = 12$ arrangements in this case.

Altogether, there are $12 \times (6 + 2 + 12) = 240$ arrangements in Case 2.

Combining Cases 1 and 2, the answer is $105 \times (57 + 240) = 31185$.

5. A number is *good* if the sum of its digits is 18, (for example 130518 is a good number), find the number of good 6-digit numbers.

Using generating functions, the number of good 6-digit numbers equals to the coefficient of z^{18} in the expansion of $A(z) = (z + z^2 + \cdots + z^9)(1 + z + z^2 + \cdots + z^9)^5$.

Now $z + z^2 + \cdots + z^9 = \frac{z(1 - z^9)}{1 - z}$ and $(1 + z + z^2 + \cdots + z^9)^5 = \left(\frac{1 - z^{10}}{1 - z}\right)^5$. Hence

$A(z) = z(1 - z^9)(1 - z^{10})^5(1 - z)^{-6}$. But

$z(1 - z^9)(1 - z^{10})^5 = z(1 - z^9)(1 - C_1^5 z^{10} + \cdots) = z - z^{10} - 5z^{11} + \cdots$ and

$(1 - z)^{-6} = 1 + C_1^6 z + C_2^7 z^2 + C_3^8 z^3 + \cdots$.

Therefore the coefficient of z^{18} in the expansion of $A(z)$ is

$= C_{17}^{22} - C_8^{13} - 5C_7^{12} = 21087$, which equals the number of good numbers.

(Alternative Solution. There are C_r^{n+r-1} ways placing r indistinguishable balls into n

distinguishable boxes. Indeed imagine the r balls and $n-1$ partitions separating the balls (excluding the two ends of the boxes) lining together and there are $(n+r-1)!$ ways arranging them, however the balls may be permuted among them and the partitions may be permuted among them and the partition will still be the same,

hence indeed there are $\frac{(n+r-1)!}{r!(n-1)!} = C_r^{n+r-1}$ ways of placing the balls into the boxes.

Now our original problem is equivalent to placing 18 balls into 6 distinguishable boxes, with the first box contains at least one ball and no balls contain more than 9 balls. We first put a ball into the first box. There are 17 balls to be distributed and

there are $C_{17}^{6+17-1} = C_5^{22}$ ways of doing this. We have to exclude the case that the first

box contains 10 balls or more, so we imagine 10 balls already be put into first box, the

remaining 8 balls are to be distributed among the 6 boxes, there are $C_8^{6+8-1} = C_5^{13}$

ways to do it. For the second to sixth boxes, we imagine 10 balls already in one box, 1 ball already in the first box, the remaining 7 balls are to be distributed among the six

boxes, and there are $C_7^{6+7-1} = C_5^{12}$ ways to do it. As these case are to be ruled out, we

have $C_5^{22} - C_5^{13} - 5 \times C_5^{12} = 21087$.)

Medium

1. Let X be a randomly chosen subset of $\{1, 2, \dots, 2017\}$. A positive integer n is said to be 'good' if both n and $n + |X|$ are elements of X (where $|X|$ denotes the number of elements of X). Find the expected value of the number of 'good' integers.

Suppose n is a card for X and X has k elements. Then $n, n + k \in X$ and so $k \geq 2$ and $k \leq 2017 - n$. The remaining $k - 2$ elements of X may be chosen from the remaining 2015 elements of $\{1, 2, \dots, 2017\}$, which gives C_{k-2}^{2015} possible sets X . Now sum over all admissible values of k and to obtain

$$C_0^{2015} + C_1^{2015} + C_2^{2015} + \dots + C_{2015-n}^{2015}$$

sets with n as a card. Sum again over all $n = 1, 2, \dots, 2015$ (since there are no admissible values of k when $n > 2015$) to be obtain a total sum S of

$$2015 C_0^{2015} + 2014 C_1^{2015} + 2013 C_2^{2015} + \dots + C_{2014}^{2015}.$$

By symmetry,

$$S = C_1^{2015} + 2 C_2^{2015} + 3 C_3^{2015} + \dots + 2015 C_{2015}^{2015}$$

and so

$$2S = 2015 (C_0^{2015} + C_1^{2015} + C_2^{2015} + \dots + C_{2015}^{2015}) = 2015 \cdot 2^{2015}.$$

Therefore the average number of cards is $\frac{2015 \cdot 2^{2014}}{2^{2017}} = \frac{2015}{8}$.

2. A robot from starts from $(0,0)$, it moves up, down, left or right a unit length each time. How many ways from $(0,0)$ to $(0,0)$ in $2n$ steps?

($n=1, 4$; $n=2, 36$; $n=3, 20^2$; in general, $(C_n^{2n})^2$. How to think? Surprise, it

should be $\sum_{m=0}^n C_m^{2n} C_m^{2n-m} C_{m-n}^{2n-m}$, why? $(C_n^{2n})^2$? Can think as going choosing from $2n$

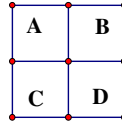
steps n steps going NE or NW, if both, then up, NE and SE, then right, etc.) Alternate: answer is

$$\begin{aligned} \sum_{k=0}^n \binom{2n}{k, k, n-k, n-k} &= \sum_{k=0}^n \frac{(2n)!}{k!k!(n-k)!(n-k)!} = \sum_{k=0}^n \frac{(2n)!n!n!}{n!n!k!k!(n-k)!(n-k)!} \\ &= \binom{2n}{n} \sum_{k=0}^n \frac{n!n!}{k!k!(n-k)!(n-k)!} = \binom{2n}{n} \sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}^2. \end{aligned}$$

3. Let n be a positive integer and suppose a $2n \times 2n$ grid is given. Suppose we color exactly $2n^2$ square boxes of the grid in such a way that the following condition is satisfied: If a box is colored then none of the boxes which share only a vertex with that box is not colored. How many ways to color the grid? Two configurations are distinct even if one can be transformed to the other by rotation or reflection.

(2012 JPN Mathematical Olympiad first stage, $(C_n^{2n})^2$ again. Consider consecutive 2×2 blocks (from top left to bottom right) and note exactly two squares of each block should be colored, then consider what happen to consecutive blocks. Finally for row blocks $\binom{2n}{n}$ choices, so are for the column blocks!

Indeed for a block:



Note exactly one of A or D should be colored and note exactly one of B or C should be colored, but how to color A or D is independent of how to color B or C. Now for the first n blocks, if D is colored, then every subsequent block (to the right) D will be colored (cannot color A). Hence we define a_1 as all the $n - a_1$ blocks A is colored and the other a_1 blocks D is colored. Correspondingly we define a_2 , etc. But note $a_1 \leq a_2$ by looking at columns. Hence

$$0 \leq a_1 \leq a_2 \leq \dots \leq a_n \leq n.$$

This corresponds to

$$1 \leq b_1 < b_2 < \dots < b_n \leq 2n,$$

where $b_t = a_t + t$, $1 \leq t \leq n$. (Altogether $\binom{2n}{n}$ choices.) Do the same for coloring B or C.)

4. Let $1 \leq r \leq n$ and consider all subsets of r elements of the set $[n] = \{1, 2, \dots, n\}$. Each of this subsets has a smallest member. Let $F(n, r)$ be the arithmetic mean of these smallest numbers. Show that $F(n, r) = \frac{n+1}{r+1}$.

(IMO1981. There are C_r^n subsets with r elements. The smallest element can be $1, 2, \dots, n-r+1$. Among them, there are C_{r-1}^{n-1} subsets with smallest element 1, there are C_{r-1}^{n-2} subsets with smallest element 2, ..., there are C_{r-1}^{n-k} subsets with smallest element k ($1 \leq k \leq n-r+1$). Thus $F(n, r) = \frac{1}{C_r^n} \sum_{k=1}^{n-r+1} k C_{r-1}^{n-k}$. Using the relation

$$C_{r+1}^{r+k+1} = C_r^r + C_r^{r+1} \dots + C_r^{r+k}, \text{ simplify to get the expression equals } C_{r+1}^{n+1} \text{ and hence the result.)}$$

The easiest solution so far known. We have the sum of these members be

$F(n, r) \binom{n}{r}$. Consider the subsets of size $r+1$ of $\{0, 1, 2, \dots, n\}$, send the subset $\{a_1, a_2, \dots, a_{r+1}\}$, with $a_1 < a_2 < \dots < a_{r+1}$, to the subset $\{a_2, \dots, a_{r+1}\}$ (of size r). Note that exactly a_2 subsets $\{i, a_2, \dots, a_{r+1}\}$, with $0 \leq i \leq a_2 - 1$, will be sent to $\{a_2, \dots, a_{r+1}\}$. Thus these group of subsets actually give the value a_2 . Together we have $\binom{n+1}{r+1} = F(n, r) \binom{n}{r}$. Simplify to get $F(n, r) = \frac{n+1}{r+1}$. Done.

5. Denote by $P_n(k)$ the number of permutations of $1, 2, \dots, n$ which have k fixed points. Show that $\sum_{k=0}^n k P_n(k) = n!$.

(For the permutation with k fixed points, then the fixed points can be chosen C_k^n

ways and other points not fixed. Thus $P_n(k) = C_k^n P_{n-k}(0)$. Hence

$$\begin{aligned} \sum_{k=0}^n k P_n(k) &= \sum_{k=1}^n k C_k^n P_{n-k}(0) = \sum_{k=1}^n n C_{k-1}^{n-1} P_{n-k}(0) \\ &= n \sum_{k'=0}^{n-1} C_{k'}^{n-1} P_{n-k'-1}(0) = n \sum_{k'=0}^{n-1} P_{n-1}(k'). \end{aligned}$$

But the final sum is $(n-1)!$.

(Double Count) Construct a rectangular table with $n!$ rows and n columns. The $n!$ rows correspond to all possible permutations of the set $\{1, 2, \dots, n\}$ and the n columns correspond to the elements of the set $\{1, 2, \dots, n\}$. We write 1 at the $i \times j$ entry of the table if j is a fixed point of the permutation i , and 0 otherwise.

Now count the number of 1s by rows. There are $p_n(0)$ rows with no 1s, $p_n(1)$ rows with one 1s, etc. Hence altogether the number of 1s is $\sum_{k=0}^n k p_n(k)$. Now for a column j ,

there will be a 1 if the corresponding permutation fixes j . Of course the number of permutations fixing j is $(n-1)!$, (it does not matter if a permutation fixes other entry or not). Hence the number of 1s, counting by columns, is $n \times (n-1)! = n!$.

(Probability argument) The expected number of fixed points of a random permutation of $\{1, 2, \dots, n\}$ is 1. This is obvious since each of the n numbers is fixed with probability $\frac{(n-1)!}{n!} = \frac{1}{n}$.

(Inclusion-Exclusion Principle, IEP) Denote by D_n the number of permutations of $\{1, 2, \dots, n\}$ that have no fixed points. Using the IEP, one show that

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right). \text{ Now}$$

$$kp_n(k) = k \binom{n}{k} D_{n-k} = n \binom{n-1}{k-1} D_{n-1-(k-1)} = np_{n-1}(k-1) \quad \text{for all } k \geq 1. \quad \text{Moreover}$$

$$D_n = p_n(0) \quad \text{and} \quad \sum_{k=0}^n p_n(k) = n!. \quad \text{Hence}$$

$$\sum_{k=0}^n kp_n(k) = \sum_{k=1}^n kp_n(k) = \sum_{k=1}^n np_{n-1}(k-1) = n \sum_{k=0}^{n-1} p_{n-1}(k) = n(n-1)! = n!.$$

(Note that the formula of D_n was never used.)

5a. (Variation of 5) For a pair of integer n and i , with $i < n$, we have

$$\sum_{k=0}^n P_i^k S(n, k) = n!. \quad (\text{Here } P_i^k = i! C_i^k \text{ if } k \geq i \text{ and } = 0 \text{ if } k < i).$$

(Put $i = 1$ to get 5.)

Indeed for each i , the number of permutations with at least i fixed points is

$$C_i^n (n-i)!. \quad \text{But on the other hand, this number is the same as summing over all } k,$$

with $i \leq k \leq n$, of the number of permutations having exactly k fixed points times the number of specified i fixed points among those k fixed points, hence

$$\sum_{k=i}^n C_i^k S(n, k) = C_i^n (n-i)!.$$

$$\text{This implies } \sum_{k=i}^n P_i^k S(n, k) = \sum_{k=0}^n P_i^k S(n, k) = n!.$$

6. (Soviet Union Mathematical Olympiad 1996) 16000 committees were formed with members coming from 1600 councillors. Each committee consisted of exactly 80 members. Show that there exist two committees, and they had at least 4 common members.

In this problem we do not consider pairs of members in each committee. Instead we consider from the perspective an individual councillor, and see how many pairs of committees he joined. Suppose councillors 1, 2, ..., 1600 joined respectively $k_1, k_2, \dots, k_{1600}$ committees. Hence altogether there are $C_2^{k_1} + C_2^{k_2} + \dots + C_2^{k_{1600}}$ "committee pairs". Now there are N committees, ($N = 16000$), then $k_1 + k_2 + \dots + k_{1600} = 80N$. We estimate these committee pairs

$$\begin{aligned} C_2^{k_1} + C_2^{k_2} + \dots + C_2^{k_{1600}} &= \frac{k_1(k_1-1)}{2} + \frac{k_2(k_2-1)}{2} + \dots + \frac{k_{1600}(k_{1600}-1)}{2} \\ &= \frac{k_1^2 + k_2^2 + \dots + k_{1600}^2}{2} - \frac{k_1 + k_2 + \dots + k_{1600}}{2} \geq \frac{1}{2} \frac{(k_1 + k_2 + \dots + k_{1600})^2}{1600} - \frac{k_1 + k_2 + \dots + k_{1600}}{2} \\ &= \frac{1}{2} \frac{(80N)^2}{1600} - \frac{80N}{2} = 2N^2 - 40N = 2N(N - 20) \end{aligned}$$

Suppose now every two committee have at most three common members, then can only be at most $3C_2^N = \frac{3N(N-1)}{2}$ committee pairs. Hence

$$2N(N - 20) \leq \frac{3}{2} N(N - 1), \text{ or } N \leq 77, \text{ contradicting } N = 16000. \square$$

(Note that in the estimate we have made use of the Cauchy-Schwarz Inequality.)

7. (International Mathematical Olympiad 1998) In a contest there are m candidates and n judges, where $n \geq 3$ is an odd integer. Each candidate is evaluated by each judge as either pass or fail. Suppose that each pair of judges agrees on at most k candidates. Prove that $\frac{k}{m} \geq \frac{n-1}{2n}$.

The problem has reminded us to consider pairs of judges who agree on particular candidates. Consider candidate i , $1 \leq i \leq m$, suppose x_i judges consider him pass and y_i judges think he fail. Then the number of pairs of judges that agree on this candidate is

$$\begin{aligned} C_2^{x_i} + C_2^{y_i} &= \frac{x_i(x_i-1)}{2} + \frac{y_i(y_i-1)}{2} = \frac{x_i^2 + y_i^2}{2} - \frac{x_i + y_i}{2} \geq \frac{1}{2} \frac{(x_i + y_i)^2}{2} - \frac{x_i + y_i}{2} \\ &= \frac{1}{4} n^2 - \frac{n}{2} = \frac{1}{4} [(n-1)^2 - 1]. \end{aligned}$$

Now n is odd and $C_2^{x_i} + C_2^{y_i}$ is an integer, hence $C_2^{x_i} + C_2^{y_i}$ is at least $\frac{1}{4}(n-1)^2$.

Also there are n judges and any pair of judges agree on at most k candidates, there are

at most kC_2^n pairs that agree on certain candidates. Thus

$$kC_2^n \geq \sum_{i=1}^m [C_2^{x_i} + C_2^{y_i}] \geq \frac{m(n-1)^2}{4}, \text{ simplify to obtain the desired result. } \square$$

Notes: We now consider a problem of slightly different flavor. Consider sequences of length n and such that each sequence contains only 0s or 1s. Define the “distance” between two such sequences as the number of positions they differ. For example 1101011 and 1011000 are two sequences of length 7, and they differ on positions 2, 3, 6 and 7, hence the distance between them is 4. Using set language, we say that a set contains 7 elements 1, 2, 3, 4, 5, 6 and 7, as sequence 1 are non-zero at positions 1, 2, 4, 6, and 7, and we may consider sequence 1 as a subset containing elements 1, 2, 4, 6 and 7. Likewise sequence 2 is a subset containing 1, 3 and 4. Elements belong to sequence 1 or sequence 2, but not both, in this case consist of 2, 3, 6 and 7, together form a subset, called the “symmetric difference” of the two subsets. One can verifies easily that the distance between the sequences corresponds precisely to the number of elements in the symmetric difference. Now our concern is, given n and a restriction on the distances between the sequences, at most how many sequences we can get?

8. Given m sequences of length n and the sequences contain only 0s and 1s. Suppose the distance between any two such sequences is at least d , then $m \leq \frac{2d}{2d-n}$.

Consider any two sequences and the corresponding pairs that differ. First we note that there are C_2^m pairs of sequences. For any pair of sequences, the number of pairs that differ or “distance” between them is at least d . Hence the sum of total distances between all possible pairs of sequences is at least dC_2^m . Now we list all of these m sequences as m rows. Then each column j , $1 \leq j \leq n$, correspond to the j^{th} position of these m sequence. If the j^{th} column contains x_j 0s, then it will contain $m - x_j$ 1s.

This column will contribute $x_j(m - x_j)$ pairs of positions that the sequences differ.

$$\text{Observe } x_j(m - x_j) \leq \left[\frac{x_j + (m - x_j)}{2} \right]^2 = \frac{m^2}{4},$$

$$\text{thus } dC_2^m \leq \sum_{j=1}^n x_j(m - x_j) \leq \sum_{j=1}^n \frac{m^2}{4} = \frac{nm^2}{4}. \text{ Simplify to get } m \leq \frac{2d}{2d-n}. \square$$

9. Let k be a positive integer. Find the number of non-negative integers n less than or

equal to 10^k

satisfying the following conditions:

- (i) n is divisible by 3;
- (ii) Each decimal digit of n is one of the digits 2, 0, 1 or 7.

As 10^k is not divisible by 3, we have to count the numbers with no more than k digits, from 0 to $\underbrace{99\dots9}_k$. Indeed we count the numbers of form $\overline{a_1a_2\dots a_k}$, with

$a_i = 2, 0, 1$ or 7 . But 3 divides $\overline{a_1a_2\dots a_k}$ if and only if 3 divides $a_1 + a_2 + \dots + a_k$, hence we have to count those tuples $(a_1, a_2, \dots, a_k)$ such that 3 divides $a_1 + a_2 + \dots + a_k$.

Define $P(x) = (x^2 + 1 + x + x^7)^k$. Then

$$P(x) = (x^2 + 1 + x + x^7)^k = \sum x^{a_1 + a_2 + \dots + a_k}.$$

The number of tuples $(a_1, a_2, \dots, a_k)$ such that 3 divides $a_1 + a_2 + \dots + a_k$ is the sum of coefficients of those terms of $P(x)$ whose corresponding degrees are divisible by 3. Let the required number of T . Now let ξ be a complex cube root of 1, we have then $\xi^3 = 1$ and $\xi^2 + \xi + 1 = 0$. Note also that if q is a multiple of 3, then $\xi^{2q} + \xi^q + 1 = 3$. But if q is not a multiple of 3, then $\xi^{2q} + \xi^q + 1 = 0$. Using this observation, we have $T = \frac{P(1) + P(\xi) + P(\xi^2)}{3}$. But from the definition of $P(x)$, we

have

$$T = \frac{P(1) + P(\xi) + P(\xi^2)}{3} = \frac{4^k + \xi^k + \xi^{2k}}{3} = \begin{cases} \frac{4^k + 2}{3}, & \text{if } 3|k \\ \frac{4^k - 1}{3}, & \text{otherwise.} \end{cases}.$$

(Alternative Solution by Lee Ker Yang, sketched) Denote by a_k, b_k, c_k the numbers of integers with k or fewer digits, with digit sums equal 0, 1 or 2 (mod 3) respectively, and those integers contain only the digits 2, 0, 1 or 7 only. Note $1 \equiv 7 \pmod{3}$.

To count a_{k+1} , we add 0 to a sequence with digit sum 0 (mod 3), or 2 to a sequence with digit sum 1 (mod 3), or 1, 7 to a sequence with digit sum 2 (mod 3). Thus

$$a_{k+1} = a_k + b_k + 2c_k.$$

Likewise

$$b_{k+1} = 2a_k + b_k + c_k,$$

$$c_{k+1} = a_k + 2b_k + c_k.$$

We also have

$$\begin{aligned}a_1 &= 1, b_1 = 2, c_1 = 1, \\a_2 &= 5, b_2 = 5, c_1 = 6, \\a_3 &= 22, b_3 = 21, c_3 = 21.\end{aligned}$$

Note if we let $d_k = a_k + b_k + c_k$, then $d_k = 4^k$ (all possible integers with digits 2, 0, 1 or 7).

We may prove by induction that for $k = 3m+1$, $a_k = c_k = \left\lfloor \frac{4^k}{3} \right\rfloor, b_k = \left\lceil \frac{4^k}{3} \right\rceil$; for

$$k = 3m+2, \quad a_k = b_k = \left\lfloor \frac{4^k}{3} \right\rfloor, c_k = \left\lceil \frac{4^k}{3} \right\rceil; \quad \text{for } k = 3m, \quad a_k = \left\lceil \frac{4^k}{3} \right\rceil, b_k = c_k = \left\lfloor \frac{4^k}{3} \right\rfloor.$$

Hence $a_k = \left\lfloor \frac{4^k}{3} \right\rfloor = \frac{4^k - 1}{3}$, when $3 \nmid k$ and $a_k = \left\lceil \frac{4^k}{3} \right\rceil = \frac{4^k + 2}{3}$, when $3 \mid k$. One

may verify these statements are consistent with the original proof.

10. (Extra). Exactly half (100 of them) of the integers 1, 2, ..., 200 are selected and their sum taken, then subtract the sum of the remaining half of the integers. The number is squared and is called a *score*. What is the average of all possible scores?

(Japan Math Olympiad 2016). Define

$$S(x_1, x_2, \dots, x_n) = \sum (\pm x_1 \pm x_2 \pm \dots \pm x_n)^2,$$

where the sum is taken over all possible choices of $\pm$ signs in which exactly n of them are + signs and the other n of them are - signs, altogether $\binom{2n}{n}$ terms in the

sum. We want to find $\frac{S(1, 2, \dots, 2n)}{\binom{2n}{n}}$, with $n=100$. Now $S(x_1, x_2, \dots, x_n)$ is

homogeneous, symmetric of degree 2 with respect to the variables $x_1, x_2, \dots, x_{2n}$.

Hence $\frac{S(x_1, x_2, \dots, x_{2n})}{\binom{2n}{n}} = a(x_1^2 + x_2^2 + \dots + x_{2n}^2) + b(x_1 + x_2 + \dots + x_{2n})^2$. Note

(i) The coefficient of each term x_j^2 is 1 for each j is 1 in each term of the sum S .

(ii) Each of the term is 0 when $x_1 = x_2 = \dots = x_{2n} = 1$.

Hence $a + b = 1$ and $0 = a + 2nb$, get $b = -\frac{1}{2n-1}$ and $a = \frac{2n}{2n-1}$. Also

$1 + 2 + \dots + 2n = n(2n+1)$ and $1^2 + 2^2 + \dots + (2n)^2 = \frac{n(2n+1)(4n+1)}{3}$. Simplify to get

$$\frac{S(1, 2, \dots, 2n)}{\binom{2n}{n}} = \frac{n^2(2n+1)}{3}. \text{ Put } n=100 \text{ to get } 670000.$$

Hard

1. (IMO1989 No. 6) Let n be a positive integer. A permutation $\{x_1, x_2, \dots, x_{2n}\}$ of the set $\{1, 2, \dots, 2n\}$ has the property P if $|x_i - x_{i+1}| = n$ for at least one i . Show that for every n , $n \geq 1$, the number of permutations with the property P is greater than the number of permutations which don't have.

Consider the set permutations, let

$A = \{(x_1, x_2, \dots, x_{2n}) \mid (x_1, x_2, \dots, x_{2n}) \text{ has property } P\}$,

$B = \{(x_1, x_2, \dots, x_{2n}) \mid (x_1, x_2, \dots, x_{2n}) \text{ doesn't have property } P\}$,

$C = \{(x_1, x_2, \dots, x_{2n}) \mid (x_1, x_2, \dots, x_{2n}) \text{ has exactly one } i \text{ such that } |x_i - x_{i+1}| = n \text{ and } i \neq 1\}$.

Note that $C \subseteq A$ and $(n+1, 1, 2, \dots, n, n+2, \dots, 2n)$ is in A but not in C . Hence

$|C| < |A|$. Now for any $y = (y_1, y_2, \dots, y_{2n}) \in B$, we have $|y_2 - y_1| \neq n$. There exists a

unique $k, k > 2$, such that the difference between y_1 and y_k is n . Put y_1 on the

left of y_k , get $y' = (y_2, \dots, y_{k-1}, y_1, y_k, \dots, y_{2n})$, a permutation of C . Easy to check

$f(y) = y'$ is a bijection from B to C . Hence finally $|C| = |B| < |A|$.

Note: A *permutation of the numbers* 1, 2, ..., n is a re-arrangement of the numbers without omission or repetition. It is easy to see there are $n!$ permutations. We may write a permutation in the form of a matrix, say

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 4 & 6 & 2 & 5 & 1 & 3 & 7 \end{pmatrix}.$$

In this way the arrangement of 1, 2, 3, 4, 5, 6 and 7 is 4, 6, 2, 5, 1, 3 and 7. We have also say 1 goes to 4, 4, goes to 5, 5 goes to 1, etc and we can write the permutation as a product of *cycles* (1, 4, 5)(2, 6, 3)(7). Here 7 is a *fixed point*.

Final Notes: Counting by exhaustion, Induction, Double Count, Generating Function, Mapping etc.

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